# LAUDS DATA RIDE
# DATA COLLECTION AND ANALYSIS WORKFLOW

## Project Overview

The Data Ride campaign was conducted in Nancy as part of the LAUDS Factories programme to explore how citizen-generated cycling data can support discussions about cycling conditions and urban mobility. The reported campaign period was 11 February to 20 May 2026.

Using open-source sensor boxes mounted on bicycles, local cyclists collected information about:

- Overtaking distance
- Lateral cycling corridors
- Cycling speed
- Road surface conditions
- Air quality
- GPS position

The campaign combined citizen science, open-source technology and participatory urban analysis. This methodology distinguishes the operations implemented in the three Python scripts from subsequent spatial processing, mapping and interpretation.

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## 1 Dataset column description

Each row represents one standardized sensor observation. The table describes the main fields; the exact exported schema depends on the processing output version.

| Column | Type or unit | Description |
|---|---|---|
| timestamp | Date and time UTC | Original UTC timestamp where retained. |
| timestamp_qgis | Date and time UTC | Simplified UTC timestamp without milliseconds or a visible timezone suffix. This is not a conversion to local time. |
| lat and lon | Decimal degrees | Latitude and longitude in WGS 84. |

| Column | Type or unit | Description |
|---|---|---|
| box_code | Text | Device code used to group observations along each device timeline. |
| anonymized_box_id | Integer | Pseudonymous numeric device identifier. Pseudonymization does not make trajectories anonymous. |
| trip_id | Text | Trip identifier constructed using gaps greater than five minutes within each device timeline. |
| direction | Text | Outbound or return assigned using geographic displacement. Not verified journey purpose. |
| distance_along_trip_m | Metres | Cumulative GPS distance within each retained trip after trimming. |
| distance_left | Centimetres | Distance to the nearest detected object on the left side. |
| distance_right | Centimetres | Distance to the nearest detected object on the right side. |
| corridor_width | Centimetres | Sum of left and right distances; missing when either input is missing. |
| finedust_pm1<br>finedust_pm2.5<br>finedust_pm4<br>finedust_pm10 | µg/m³ | Mobile particulate matter measurements for the corresponding particle-size fractions. |
| overtaking_manoeuvre | Score 0 to 1 | On-device model score used as an input to subsequent event selection. |
| speed | Metres per second | Recorded speed. Multiply by 3.6 to express it in km/h. |

| Column | Type or unit | Description |
|---|---|---|
| standing | Score 0 to 1 | Indicator of whether the bicycle is stationary. |
| roughness_score | Score 0 to 1 | Normalized weighted indicator calculated from the surface classification scores. |

Distance readings can include vehicles, walls, vegetation and other objects. Low distance alone does not establish an overtaking event. PM readings are mobile campaign observations, not official annual monitoring values.

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## 2 Data Collection

The reported campaign period is 11 February to 20 May 2026. The updated download script requests 1 February to 21 May 2026 at 23:59:59 UTC, and the supplied standardized dataset contains observations from 11 February to 21 May. The reporting period and the available data period therefore differ by one day at the end; the scripts do not impose the official campaign end date during standardization.

The supplied standardized dataset contains 193,328 observations, 357 trip identifiers and 11 distinct device identifiers. Project documentation reports 10 participants. Participant counts and device counts must be reported separately; the reason for the additional device identifier has not been established.

The presentation additionally reports approximately 1,234.74 km and 84 recorded hours. Their GPS-jump exclusions and duration calculations are not implemented or fully specified by the three scripts.

| Sensor or source | Variable |
|---|---|
| Smartphone GPS | Position and speed |
| Left Time of Flight sensor | Left lateral distance and input to the overtaking model |
| Right Time of Flight sensor | Right lateral distance |
| Particulate matter sensor | PM1, PM2.5, PM4 and PM10 |
| Motion sensing and classification | Surface and stationary indicators |

# 3 Data Transmission

Measurements were uploaded to openSenseMap through the connected smartphone workflow. They were subsequently downloaded through the API for processing outside the device.

# 4 Data Extraction

Script 00 requests measurements in three-day blocks and exports long and wide CSV files. The updated date range extends through 21 May 2026. Updating the date configuration does not, by itself, demonstrate that a new download exactly reproduces the historical input files; the supplied download manifest also records a complementary download in May.

For a reproducible run, keep the download manifest and use the intended wide-format files as inputs. Script 01 reads CSV files from its input directory, so mixing long and wide exports or overlapping download periods can introduce unintended inputs. Duplicate handling at block boundaries must be checked rather than assumed.

The supplied processing files are 00_download_boxes.py, 01_standardization_cleaning.py and 02_overtaking_event_detection.py. Their principal outputs are outcome_01_standardized.csv, overtaking_candidates_debug.csv and overtaking_events_final.csv.

# 5 Data Cleaning

Timestamps are parsed in UTC. Invalid timestamps are dropped, numeric fields are standardized, and observations with missing or out-of-range coordinates are removed. Valid coordinates are latitude from −90 to 90 degrees and longitude from −180 to 180 degrees. Rows missing all of distance_left, distance_right, overtaking_manoeuvre, speed and standing are also removed.

Observations are sorted along each device timeline. A gap strictly greater than 300 seconds starts a new trip. The first and last 15 seconds of each trip are excluded. Cumulative distance is then recalculated from consecutive GPS positions using the haversine formula, starting at zero for each retained trip.

### Trip direction

The direction field assigns outbound or return from the dominant geographic displacement between the trip endpoints. These are heuristic labels, not verified journey purposes or paired outward and return journeys.

### Scope of initial filters

Script 01 does not apply a general 30–400 cm left-distance filter and does not remove all speeds above 15 m/s. These filters must not be presented as implemented initial-cleaning operations. The overtaking-specific distance filter is applied separately in script 02.

# 6 Identifying Stationary Observations

Stationary observations remain in the standardized dataset so they can support stopping-location analysis. Script 02 removes observations with standing > 0.95, together with a two-second buffer before and after each such observation within the same trip. The hotspot description uses 0.95–1.00; the boundary at exactly 0.95 differs from the event-filter rule.

# 7 Overtaking Event Detection

Script 02 processes downloaded observations in Python. It is separate from the model running on the senseBox device. The on-device overtaking score is one input to event selection, together with distance variation, temporal grouping and filters for stationary or repeated patterns.

### Distance and stationary filters

Left-distance readings from 30 to 300 cm inclusive are retained. Observations with standing > 0.95 and observations within ±2 seconds of them are excluded. These filters apply to overtaking analysis, not to every map or to the entire standardized dataset.

### Candidate detection

Each trip is scanned using forward two-second windows, including both endpoints. A window must contain at least two observations. A candidate is retained when its minimum left distance is at most 250 cm and its distance variation, defined as maximum minus minimum, is from 30 to 120 cm inclusive. The observation at minimum distance represents the candidate. Duplicate candidates are removed using trip, timestamp, window minimum and variation.

### Repeated pattern removal

Consecutive candidates belong to the same group when their time gap is at most 3 seconds and their displacement along the trip is at most 20 metres. A group is removed when at least two conditions hold: it contains at least three candidates; its spatial extent is at least 15 metres; or its duration is at least 5 seconds. This is a heuristic filter for patterns resembling encounters with parked vehicles or other persistent objects, not a verified object classification.

### Event consolidation and confidence

Remaining candidates are merged within a trip when consecutive candidate timestamps are at most one second apart. Event statistics are calculated over the combined candidate windows using the filtered observation data. The representative location is the observation with the minimum left distance, recorded as event_dmin.

Only events with a maximum overtaking-manoeuvre score of at least 0.20 are retained. Scores from 0.20 to below 0.70 are labelled medium confidence; scores of at least 0.70 are labelled high confidence. These labels express pipeline criteria and should not be interpreted as calibrated probabilities that an event is real.

### Verified output and benchmark estimates

The supplied standardized table reproduced the supplied script 02 outputs exactly in the earlier comparison: 8,903 candidates in the candidate export and 1,324 retained events, comprising 835 medium-confidence and 489 high-confidence events. Of the retained events, 833 have event_dmin < 100 cm, corresponding to 62.9%.

Applying the Porto precision benchmark[1] of 47.1% gives approximately 624 likely true events overall and 392 likely true events below one metre. These are aggregate estimates calculated

[1] The Porto benchmark used a cycling video alongside sensor data recorded on 19 March 2026, covering approximately 27 minutes and 1,577 data points. Overtaking occurrences were manually identified in the video and compared with sensor-based detections. The reviewed worksheet contains 57 positively annotated reference rows. Previously reported precision and recall were 47.1% and 17.5%, respectively. This preliminary assessment evaluated event detection, not distance-measurement accuracy or the validity of the dual-sided corridor indicator.

after detection, not additional verified events or a classification of particular locations. Applying the same factor to both counts leaves the below-one-metre share unchanged.

Transfer of the Porto benchmark assumes comparable detection performance in Nancy. It does not compensate for missed overtaking events or establish the total number of actual passes during the campaign. Local video validation would be needed to assess performance in Nancy.

# 8 Corridor Indicator

corridor_width = distance_left + distance_right, in centimetres. The result is missing when either reading is missing. Its interpretation depends on the objects detected on both sides of the bicycle.

The indicator describes the sum of lateral distances to detected objects. It does not directly measure cycling-infrastructure width. For overtaking events, script 02 also exports corridor_mean and corridor_min over the event observations.

# 9 Road Surface Indicator

The roughness indicator combines surface_asphalt, surface_paving, surface_compacted and surface_sett, with weights 1, 2, 3 and 4 respectively. Let A, P, C and S denote these scores. The calculation is: roughness_score = (((A + 2P + 3C + 4S) / (A + P + C + S)) − 1) / 3, clipped to the interval 0–1.

Missing surface scores contribute zero to the weighted sum; a row without usable surface information or with a zero score sum produces a missing result. The input scores may use 0–1 or 0–100 scales when the same scale is used consistently. This script does not query OSM to calculate roughness. OSM is used later as a spatial aggregation network.

# 10 Spatial Processing

Spatial analyses were performed in QGIS. The Métropole du Grand Nancy provided the parking, cycling-infrastructure and tree datasets. The street network was obtained directly from OpenStreetMap (OSM), independently of the metropolitan datasets. Final visualizations used a kepler.gl-based interface hosted on data.atrai.bike. Sensor coordinates originate in WGS 84; subsequent local spatial processing used EPSG:2154, RGF93 / Lambert-93. The haversine calculations in script 01 use latitude and longitude.

Sensor observations and detected overtaking events were linked to the relevant spatial layers in QGIS. Nearest-neighbour spatial joins associated observations and events with OSM street segments and the local cycling-infrastructure dataset. Parking proximity analysis and tree buffers supported the corresponding map indicators. These operations were carried out separately from the three Python scripts. The maximum matching distance, tie handling, network version and unmatched-point rules are not recorded in the available documentation; these parameters are needed for exact reproduction.

# 11 Visualization and Final Maps

The following sections document the nine map themes. Spatial outputs and their presentation-reported percentages are distinguished from values reproduced by the Python event-detection workflow.

### Map 01 Rides per user

The map displays GPS coverage by device or participant and trip. It uses the standardized observations and device grouping. Device codes distinguish recording sources; they should not automatically be equated with distinct people. This map describes campaign coverage rather than a representative sample of all cycling in Nancy.

### Map 02 Standing hotspots

The map highlights the concentration of observations with standing values from 0.95 to 1.00. Such locations may reflect intersections, signals, waiting or other interruptions; the indicator alone does not identify their cause.

The presentation reports 357 trips, an average trip speed of 14.68 km/h and approximately 3.86% of recorded time stationary. The trip count agrees with the supplied table. The speed-averaging method and interval-weighting rules for standing time have not been recovered. These two indicators remain presentation-reported values rather than independently reproduced results.

The presentation also reports approximately 1,234.74 km and 84 hours. It describes excluding unrealistic GPS jumps for distance, but that exclusion rule is not implemented in script 01. The distance filter and duration calculation must be documented before these totals can be reproduced.

### Map 03 Bike path conditions

In QGIS, roughness observations were assigned to OSM segments and summarized using the mean roughness for each segment. The presentation reports 442.26 km, or 88.4%, classified as smooth or moderate and 58.20 km, or 11.6%, as rough or very rough. These percentages refer to mapped OSM segment length with data, not accumulated cycling distance.

The numerical class boundaries were not supplied. The aggregate percentages and lengths are therefore reported map results, not outputs reproduced by the three scripts. Segment selection and treatment of missing roughness also require documentation.

### Map 04 Cycling speed

Speed readings are converted from m/s to km/h by multiplying by 3.6 and summarized by street segment. The presentation reports 5.8% in the below-5 km/h category, 40.4% in the 5–15 km/h category and 53.8% above 15 km/h. The supplied methodology leaves mean versus median unresolved; the percentage denominator and exact threshold inclusivity must also be specified. The map should be read as sampled cycling speed, not a direct measure of traffic congestion.

### Map 05 Corridors near on street parking

In QGIS, the analysis averaged corridor_width by OSM segment and selected segments near parking locations classified as on-street parallel parking, or créneau. The presentation reports that 23.4% of the selected mapped length, approximately 65 km, had mean corridor width below one metre. The parking layer was provided by the Métropole du Grand Nancy; street segments were obtained directly from OSM.

The parking proximity threshold is not supplied. These readings describe lateral distances to detected objects; they are not restricted to overtaking events and do not establish the presence of a parked vehicle at each observation. This result cannot be reproduced from the three scripts alone.

### Map 06 Overtaking distance

The map uses potential overtaking events from script 02 and event_dmin, the minimum lateral distance during each event. Events were subsequently matched to OSM segments in QGIS. When presenting segment summaries or street rankings, the aggregation statistic and treatment of multiple events must be stated separately.

The 62.9% share below one metre and estimate of approximately 392 likely true close passes agree with the supplied event table and subsequent benchmark calculation. The benchmark scales aggregate counts; it does not locate which individual detections are true.

### Map 07 Events by infrastructure category

In QGIS, potential overtaking event locations were joined to the cycling-infrastructure dataset provided by the Métropole du Grand Nancy. The analysis selected relevant right-side and unidirectional categories and calculated the share of events with a minimum measured lateral distance below one metre within each category. Categories include no infrastructure, painted cycle lanes, separated cycle tracks, mixed pedestrian–cycling spaces, bus and cycle lanes, and contraflow facilities.

The presentation table reports 79 estimated total events in the no-infrastructure category, including 38 below one metre. Painted cycle lanes have a reported below-one-metre share of 65.2%, the highest among the principal categories highlighted in the presentation.

The revised category estimates below one metre sum to approximately 102, including events on streets without cycling infrastructure. These provisional estimates apply the previously reported Porto precision factor of 47.1%. They do not confirm individual events or their locations. Spatial matching, category selection and unrounded totals remain subject to verification.

### Map 08 Particulate matter

In QGIS, PM2.5 observations were joined to OSM street segments and summarized using the median. The presentation reports that 19.2% of mapped street length had a segment median above 10 µg/m³. The threshold is a descriptive reference for this map.

### Map 09 Particulate matter and trees

In QGIS, a 10-metre buffer was created around each street segment obtained directly from OSM. Trees from the dataset provided by the Métropole du Grand Nancy were filtered to retain living trees, classified as VIVANT, and intersecting trees were counted within each buffer. These counts were shown with median PM2.5.

# 12 Reproducibility and Interpretation

### Information needed for exact map reproduction

Archive the exact input files, date range, script versions, dependencies and commands; identify any overlapping downloads and their handling.

Record the extraction date and version of the street network obtained directly from OpenStreetMap, and the versions of the parking, cycling-infrastructure and tree datasets provided by the Métropole du Grand Nancy, including segmentation and coordinate transformations.

Recover matching distances, unmatched-point rules, roughness class thresholds, speed aggregation, parking proximity and valid-reading filters.

Document duration and stationary-time weighting, GPS-jump exclusions, map percentage denominators and unrounded infrastructure counts.

Keep the QGIS project, processing history or equivalent spatial scripts alongside the released methodology and map outputs.

### Public data and interpretation

The processed dataset derives from measurements publicly available on openSenseMap. Device identifiers were pseudonymized during processing. The dataset retains timestamps and geographic coordinates to support reproducibility and spatial analysis and should therefore not be described as fully anonymized.

The findings combine sensor observations, spatial processing and interpretation informed by participatory activities. They describe the sampled routes and conditions during the campaign. Policy recommendations should distinguish measured findings from participant experiences and proposed actions.